

ACUTE ABDOMEN — Clinical Revision Table (Dr Farman O Hassan)

Bold = high-yield. Acute abdomen = sudden, spontaneous, non-traumatic, severe abdominal pain, usually <24h.

1 — EPIDEMIOLOGY & CAUSES BY SYSTEM

Epidemiology	~10% of ED visits; ~40% need surgery (~1/3 if >65y). Young → appendicitis . Older → biliary, obstruction, ischaemia, diverticulitis
Intestinal	Appendicitis, mesenteric adenitis, Meckel's, perforated PU, gastroenteritis, diverticulitis, obstruction, strangulated hernia
Hepatobiliary / Vascular	HB: biliary colic, cholecystitis, cholangitis, pancreatitis, hepatitis. Vascular: ruptured AAA, acute mesenteric ischaemia, ischaemic colitis
Urological / Gynae	Uro: renal colic, UTI, testicular torsion, retention. Gynae (ECTOPIC): Ectopic, endometriosis, Cyst rupture, Torsion ovary, Ovulation (Mittelschmerz), PID, Incomplete abortion, Cystitis
Medical (non-surgical)	Metabolic: uraemia, DKA, Addisonian crisis, porphyria, FMF. Haem: sickle crisis, leukaemia. Toxins: lead, narcotic withdrawal. Referred: MI, pericarditis, pneumonia, PE → ALWAYS ECG to exclude MI

2 — PAIN PATHOPHYSIOLOGY

Type	Features	Innervation
Visceral	Midline, vague, deep, dull, poorly localized, cramping. Triggered by inflammation/ischaemia/distension	Bilateral autonomic
Parietal (somatic)	Sharp, severe, well-localized, lateralizes to quadrant. Worse with movement/cough, eased lying still	Unilateral spinal somatic
Embryological → midline site	Foregut → epigastric (stomach, pancreas, liver, biliary). Midgut → periumbilical (early appendicitis). Hindgut → suprapubic	—

3 — HISTORY: ONSET / CHARACTER / VOMIT

Sudden (seconds)	Perforated viscus, ruptured AAA/ectopic, infarction/embolism (mesenteric, MI)
Rapid (minutes)	Colic (biliary, ureteric), inflammation (appendicitis, pancreatitis, diverticulitis), ischaemia, obstruction
Gradual (hrs-days)	Neoplasm, chronic inflammation, large bowel obstruction
Migration	Periumbilical → RIF = acute appendicitis (visceral→parietal). Colicky → constant = strangulation/ischaemia → peritonitis
Character	Colicky (intermittent, pain-free) → obstruction/gastroenteritis. Burning constant → peritonitis. Severe agonizing → pancreatitis. Throbbing → cholecystitis
Vomit type	Bilious=obstruction/sepsis ; coffee-ground=gastritis/GU; fresh blood=Mallory-Weiss/GU; food=gastroenteritis/early SBO; feculent=late obstruction

4 — EXAMINATION (Inspect → Auscultate → Percuss → Palpate)

Position / vitals	Peritonitis → lies still, knees flexed. Colic → restless/rolling. Tachycardia=early shock (masked by β-blocker). qSOFA = 2 of: altered sensorium, tachypnoea, hypotension
Inspect / Auscultate	Distension, flank/periumbilical ecchymosis (pancreatitis). Absent bowel sounds = ileus/peritonitis/shock; hyperactive = obstruction/gastroenteritis; bruit = AAA
Percuss / Palpate	Loss of liver dullness = free air = perforation. Fluid wave = ascites. Rebound = peritonitis. Guarding (voluntary) vs Rigidity (involuntary = peritonism). ALWAYS do inguinal/genitalia + rectal + pelvic
Key errors	Palpate before inspect/auscultate · skip rectal/pelvic/groin · over-rely on absent rebound · miss peritonitis in immunosuppressed · forget mesenteric ischaemia (pain out of proportion)

5 — SIGNS BY CONDITION

Paralytic ileus	Distension, minimal bowel sounds, no localized tenderness
Intestinal obstruction	Distension, visible peristalsis (early)/quiet (late), diffuse pain no rebound, hernia/rectal mass
Bleeding	Pallor, shock, distension, pulsatile (AAA) or tender (ectopic) mass
Inflamed mass / abscess	Tender mass, bump tenderness, special signs (Murphy, psoas, obturator)
Perforated viscus	Scaphoid tense abdomen, ↓bowel sounds (late), loss of liver dullness , guarding/rigidity
Peritonitis	Motionless, absent bowel sounds (late), cough/rebound tenderness, guarding/rigidity

6 — INVESTIGATIONS

Bloods / urine	Neutrophil leukocytosis, CRP, ABG, S. lactate (ischaemic bowel/sepsis) , lipase/amylase (pancreatitis), LFT, β-hCG (all women childbearing age) . Urine: haematuria/pyuria, pregnancy test
Imaging	Erect CXR: subdiaphragmatic air = perforation (more sensitive than AXR; 1mL air visible). NOT usually in burst appendix. AXR: air-fluid levels, string of pearls (SBO), stones (90% renal/10% gallstone). US: best in pregnancy + gynae ; Doppler mesenteric. CT: best accuracy, unknown dx. CTA/MRA: mesenteric ischaemia + GI bleed (Dx + embolize)
Endoscopy / laparoscopy	ERCP (cholangitis), sigmoidoscopy (detorse sigmoid volvulus), colonoscopy (lower GI bleed). Laparoscopy = Dx + therapeutic; young women (ruptured follicle/PID vs appendicitis)

7 — MANAGEMENT & SURGICAL DECISION

Immediate	ABC: airway, O2 15L, large-bore cannula + IVF (N saline/Ringer's). Analgesia IV/IM. NG tube (vomiting/obstruction), Foley (hydration/urine), antibiotics (inflammation/perforation). Reassess + refer
Urgent surgery	Bleeding (ruptured AAA, ectopic, bleeding PU → shock). Physical: involuntary guarding/rigidity, spreading tenderness, progressive distension, tender mass + fever/hypotension, suspected ischaemia (acidosis/fever/tachy), deterioration. Radiology: pneumoperitoneum , free contrast leak, mesenteric occlusion. Paracentesis: blood/bile/pus/bowel/urine
When NOT to operate	Surgical (conservative): appendicular mass, pancreatitis, acute diverticulitis, cholangitis, ruptured ovarian cyst. Medical: MI/pericarditis, pneumonia, porphyria/sickle, DKA, hepatitis, pyelonephritis
NSAP	Nonspecific abdominal pain = 1/3 of cases, commonest (esp children); mild, self-resolving (viral/mild bacterial). Manage: repeat exam, imaging, gynae consult